

Q3(a)(i)

A particle is said to be undergoing free oscillations when it is oscillating at its own natural frequency.

[Accept oscillations when the only force is the restoring force; oscillations in the absence of frictional forces]

Q3(a)(ii)

Oscillations are said to be damped when there are one or more resistive forces acting on the oscillating particle.

Q3(a)(iii)

Forced oscillations occur when a system is subjected to an external force. Instead of oscillating at its natural frequency, the system will oscillate at the frequency of the external force.

Q3(b)(i)

The natural frequency of the system is approximately 35 Hz.

When the driving frequency of the driver is approximately equal to the natural frequency, the amplitude of the oscillating system will reach a maximum and resonance is said to occur.

Q3(b)(ii)

Maximum kinetic energy, $KE_{\max} = \frac{1}{2}m\omega^2x_0^2$, where $\omega = 2\pi f$

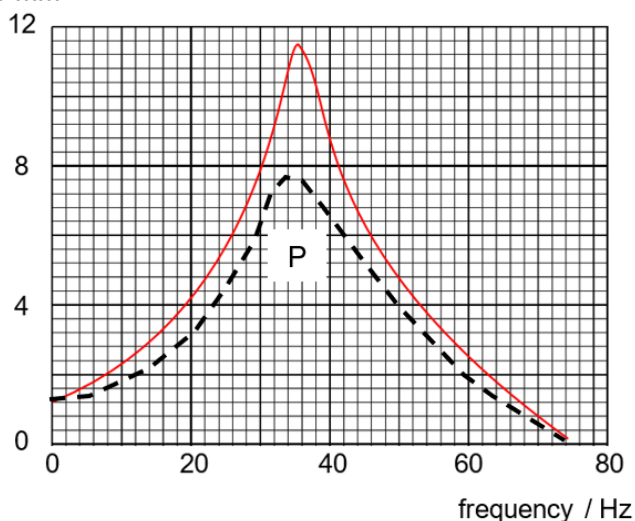
$$KE_{\max} = \frac{1}{2} \times 0.046 \times (2\pi \times 35)^2 \times (11.6 \times 10^{-3})^2$$

$$= 0.01497 \approx \mathbf{0.15 \text{ J}}$$

[Accept $11.4 \text{ mm} \leq x_0 \leq 11.6 \text{ mm}$; accept $KE_{\max} = 0.14 \text{ J}$ or 0.15 J]

Q3(b)(iii)

amplitude / mm



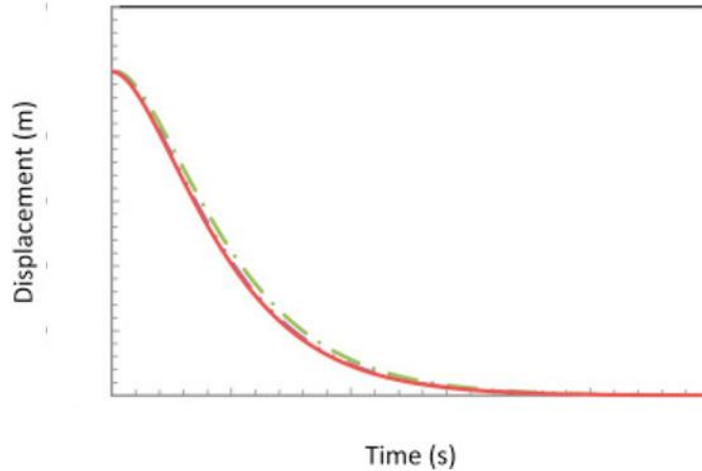
The peak of graph P is at 35 Hz or at a slightly lower frequency.

The entire graph for P must be below the original graph.

Q3(c)(i)

Critical damping occurs when the displacement decreases to zero in the shortest time, without any oscillation.

Q3(c)(ii)



Q4(a)(i)